

ASSIGNMENT -1

PCCST501 –

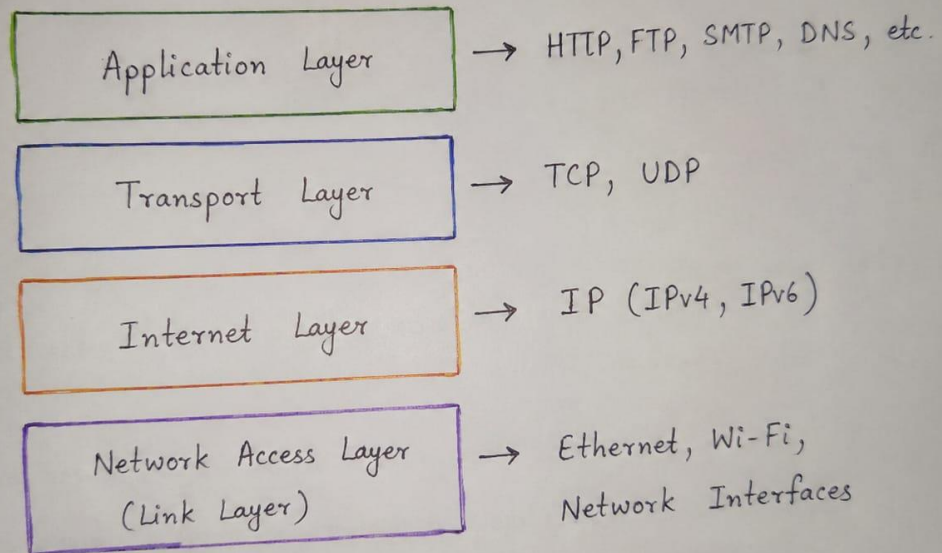
Computer Networks

Title: How the Internet Delivers a Web Page

Website Chosen: <https://www.amazon.in>

1. Draw the TCP/IP protocol stack and explain the role of each layer.

TCP/IP Protocol Stack



1. Application Layer

Provides services to users and applications. Uses protocols such as HTTP, HTTPS, DNS, FTP, and SMTP. In Amazon, this layer sends web page requests and receives webpage data.

2. Transport Layer

Ensures reliable communication. TCP guarantees error-free and ordered delivery. Divides data into segments and reassembles them.

3. Internet Layer

Uses the Internet Protocol (IP). Finds the best route for packets. Delivers packets from the client to Amazon's server.

4. Network Access Layer

Transfers data between devices on the local network. Uses Ethernet or Wi-Fi technology.

5. Physical Layer

Transmits data as electrical, optical, or wireless signals. Includes cables, fiber optics, and radio waves.

2. Explain the role of DNS in converting the domain name into an IP address

The Domain Name System (DNS) acts like the Internet's phonebook. It converts a human-readable domain name, such as www.amazon.com, into an IP address that computers use to communicate with each other.

Working of DNS

1. The user enters `www.amazon.com` in the web browser.
2. The browser first checks its local DNS cache to see if the IP address is already stored.
3. If the IP address is not found, the browser sends a DNS query to the DNS server.
4. The DNS server searches for the IP address associated with `www.amazon.com`.
5. The DNS server returns the correct IP address to the browser.
6. The browser uses this IP address to establish a connection with Amazon's web server and requests the webpage.

3. Describe how HTTP is used to request and receive the webpage.

HTTP (HyperText Transfer Protocol) is used to transfer web pages between a web browser (client) and a web server. When a user enters www.amazon.com, the browser sends an HTTP request to the Amazon server. The server processes the request and returns an HTTP response containing the webpage data (HTML, CSS, JavaScript, and images). The browser interprets this data and displays the webpage. Amazon uses HTTPS (HyperText Transfer Protocol Secure), which encrypts communication between the browser and the server for secure data transfer..

Steps in HTTP Communication

1. The user enters **www.amazon.com** in the browser.
2. The browser sends an **HTTP GET request** to the Amazon web server.
3. The web server receives and processes the request.
4. The server sends an **HTTP response** containing the webpage files.
5. The browser receives the response and renders the webpage on the screen.

4. Explain how TCP provides reliable communication for HTTP.

TCP (Transmission Control Protocol) provides reliable communication for **HTTP** by ensuring accurate, ordered, and loss-free data delivery. Before HTTP data transfer, TCP establishes a connection using the **Three-Way Handshake**. During communication, TCP detects errors, retransmits lost packets, and ensures successful delivery.

How TCP Ensures Reliable Communication

1. **Connection Establishment:** TCP creates a connection using the **Three-Way Handshake** before sending HTTP data.
2. **Reliable Data Delivery:** The receiver acknowledges every packet, and TCP retransmits lost packets.
3. **Error Detection:** TCP uses checksums to detect transmission errors.
4. **Ordered Delivery:** TCP delivers packets in the correct sequence.
5. **Flow Control:** TCP regulates the data transmission rate to avoid overwhelming the receiver.
6. **Congestion Control:** TCP reduces the sending rate during network congestion.

5. Identify whether the application follows a Client–Server or Peer-to-Peer architecture and justify your answer.

The Amazon website follows the Client–Server architecture.

In this architecture, the web browser acts as the client, while Amazon's web servers act as the servers. When a user visits **www.amazon.com**, the browser sends a request to the Amazon server. The server processes the request, retrieves the required information (such as product details, images, and prices), and sends the webpage back to the browser. The browser then displays the webpage to the user.

Unlike a **Peer-to-Peer (P2P)** network, users do not communicate directly with each other or share website data. All requests and responses are handled by Amazon's centralized servers.

6. Compare HTTP and FTP with at least four differences.

HTTP vs FTP		
Basis	HTTP	FTP
Full form	HyperText Transfer Protocol	File Transfer Protocol
Purpose	Used to access and transfer web pages	Used to transfer files between systems
Default Port	80 (HTTP), 443 (HTTPS)	21 (control), 20 (data)
Data Types	Transfers text, HTML, images, videos, etc.	Transfers all types of files
Connection	Uses a single connection	Uses two connections (control and data)
Authentication	Not required for most websites	Requires username and password
Security	Less secure (HTTP), secure with HTTPS	Less secure, password sent in clear text

7. Explain How Cookies Improve User Experience with One Real-World Example

Cookies are small text files stored by a web browser on a user's device. They help websites remember user information, making future visits faster and more personalized.

How Cookies Improve User Experience

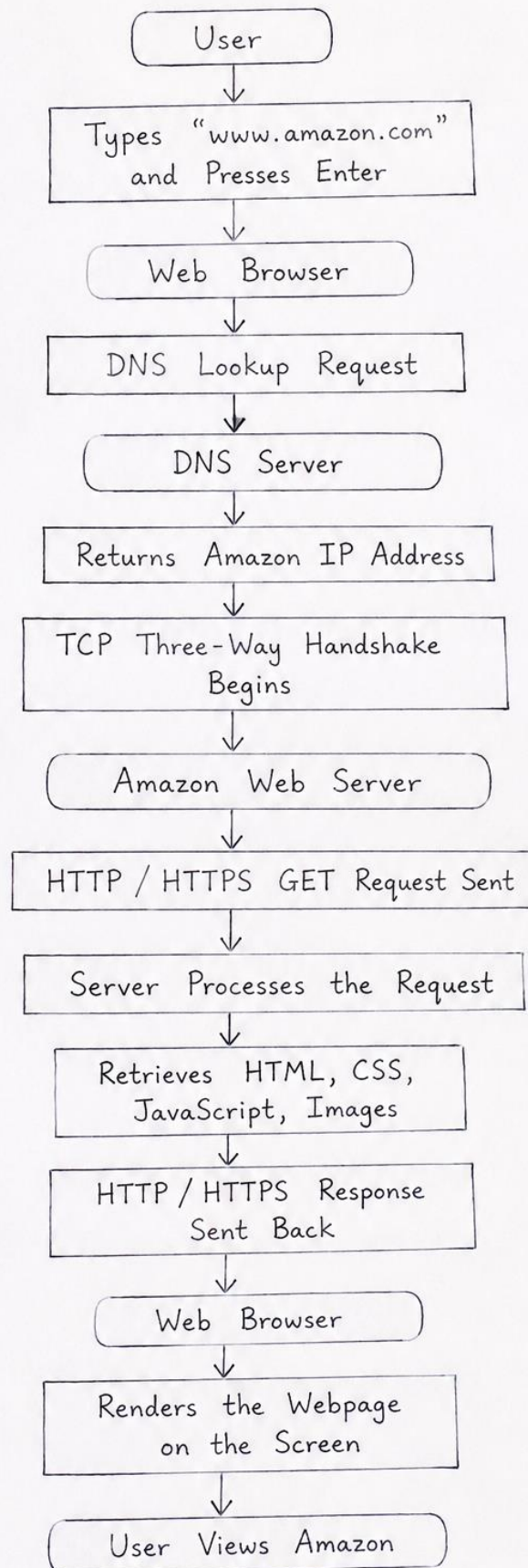
- Keep users logged in.
- Remember shopping cart items.
- Store user preferences (language and location).
- Provide personalized recommendations.
- Make browsing faster and more convenient.

Example (Amazon)

If a user adds a wireless headset to the Amazon cart but closes the browser before purchasing, Amazon stores the cart information in a cookie. When the user returns, the headset remains in the shopping cart, allowing them to continue shopping without adding it again.

8. Flow Diagram Showing the Complete Communication from the Browser to the Web Server and Back

The following flow diagram shows the complete process that occurs when a user opens **www.amazon.com** in a web browser.



Explanation

1. The user enters www.amazon.com in the web browser.
2. The browser sends a **DNS request** and receives Amazon's IP address.
3. The browser establishes a **TCP connection** using the **Three-Way Handshake**.
4. The browser sends an **HTTP/HTTPS GET request** to the Amazon server.
5. The server processes the request and retrieves the webpage files.
6. The server returns an **HTTP/HTTPS response** with **HTML, CSS, JavaScript, and images**.
7. The browser renders the data and displays the **Amazon webpage**.

9. Write a brief conclusion (100–150 words) describing the importance of applicationlayer protocols in everyday Internet usage.

Application-layer protocols are essential for enabling communication between users and Internet services. Whenever a user accesses a website like Amazon, several protocols work together to provide a smooth browsing experience. DNS converts the website name into an IP address, TCP establishes a reliable connection, and HTTP/HTTPS transfers webpage data between the browser and the web server. Cookies further enhance the user experience by remembering login details, shopping carts, and personal preferences. These protocols work behind the scenes to ensure that webpages load quickly, accurately, and securely. Without application-layer protocols, everyday online activities such as shopping, online banking, email, social media, and video streaming would not be possible. Therefore, application-layer protocols form the foundation of modern Internet communication and make online services efficient, reliable, and user-friendly.